

Control car and read encoder-USART

1.1 Explanation

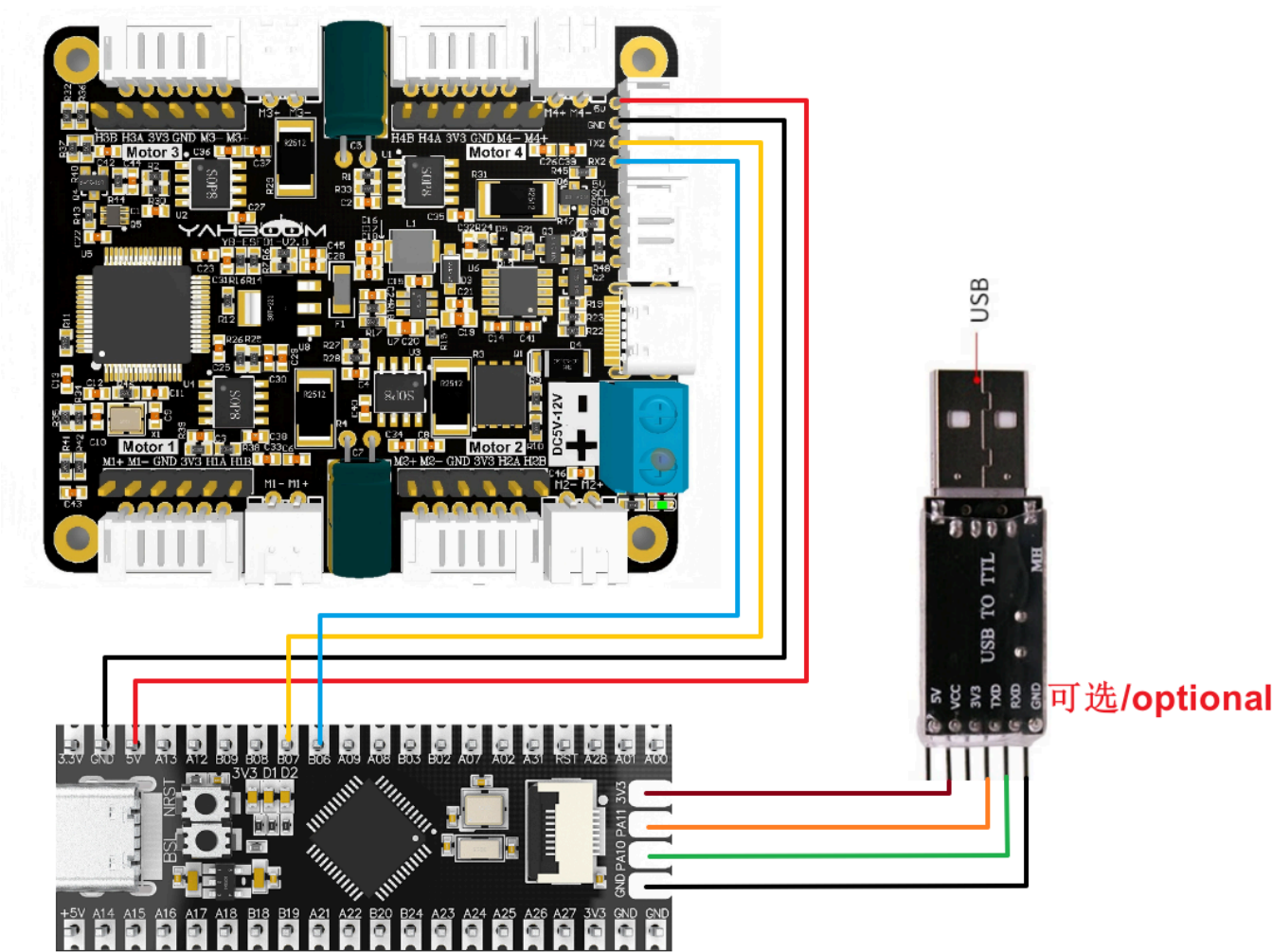
Please read 《0. Motor introduction and usage》 first to understand the motor parameters, wiring method, and power supply voltage you are currently using. To avoid improper operation and damage to the driver board or motor.

I2C and serial communication cannot be shared, only one can be selected.

The 4 motor interfaces on the module correspond to motors on robot car, as shown below:

- M1 -> Upper left motor (left front wheel of the car)
- M2 -> Lower left motor (left rear wheel of the car)
- M3 -> Upper right motor (right front wheel of the car)
- M4 -> Lower right motor (right rear wheel of the car)

Hardware wiring:



4-channel motor drive board	MSPM0G3519
RX2	PB6
TX2	PB7

4-channel motor drive board	MSPM0G3519
GND	GND
5V	5V

Motor	4-channel motor drive board(Motor)
M2	M-
V	3V3
A	H1B
B	H1A
G	GND
M1	M+

The USB to TTL serial port module is connected here to print the processed encoder data. If you are using the MSPM0G3519 purchased from our store, you can directly connect the TYPE-C interface to the computer USB port, and you can also use serial port printing, so you do not need to connect a USB to TTL serial port module.

USB TO TTL	MSPM0G3519
VCC	3V3
GND	GND
RXD	PA10
TXD	PA11

Serial port configuration: **Baud rate 115200, no parity check, no hardware flow control, 1 stop bit**

Note: The serial port here is used to print data on the serial port assistant, not for communication with the driver board

1.2 Code analysis

```
#define UPLOAD_DATA 2 // 1: Receive total encoder data 2: Receive real-time encoder data

#define MOTOR_TYPE 1 //1:520 motor 2:310 motor 3:speed code disc TT motor 4:TT DC
reduction motor 5:L type 520 motor
```

- UPLOAD_DATA: used to set the data of the motor encoder. Set 1 to the total number of encoder pulses and 2 to the real-time pulse data of 10ms.
- MOTOR_TYPE: used to set the type of motor used. Just modify the corresponding numbers according to the comments according to the motor you are currently using. You don't need to modify the rest of the code.

If you need to drive the motor and observe the data, just modify the two numbers at the beginning of the program. No changes are required to the rest of the code.

```
#if MOTOR_TYPE == 1
send_motor_type(1); //Configure motor type
delay_ms(100);
send_pulse_phase(30); //Configure the reduction ratio. Check the motor manual to find out
delay_ms(100);
send_pulse_line(11); //Configure the magnetic ring wire. Check the motor manual to get
the result.
delay_ms(100);
send_wheel_diameter(67.00); //Configure the wheel diameter and measure it
delay_ms(100);
send_motor_deadzone(1900); //Configure the motor dead zone, and the experiment shows
delay_ms(100);

#elif MOTOR_TYPE == 2
send_motor_type(2);
delay_ms(100);
send_pulse_phase(20);
delay_ms(100);
send_pulse_line(13);
delay_ms(100);
send_wheel_diameter(48.00);
delay_ms(100);
send_motor_deadzone(1600);
delay_ms(100);

#elif MOTOR_TYPE == 3
send_motor_type(3);
delay_ms(100);
send_pulse_phase(45);
delay_ms(100);
send_pulse_line(13);
delay_ms(100);
send_wheel_diameter(68.00);
delay_ms(100);
send_motor_deadzone(1250);
delay_ms(100);

#elif MOTOR_TYPE == 4
send_motor_type(4);
delay_ms(100);
send_pulse_phase(48);
delay_ms(100);
send_motor_deadzone(1000);
delay_ms(100);

#elif MOTOR_TYPE == 5
send_motor_type(1);
delay_ms(100);
send_pulse_phase(40);
delay_ms(100);
```

```

send_pulse_line(11);
delay_ms(100);
send_wheel_diameter(67.00);
delay_ms(100);
send_motor_deadzone(1900);
delay_ms(100);
#endif

```

This is used to store the parameters of the motors sold in our store. By modifying the MOTOR_TYPE parameter above, one-click configuration can be achieved. Normally, when using the motor of our store, do not modify the code here. If you are using your own motor, or if a certain data needs to be modified according to your needs, you can check the document 《1.2 Control command》 to understand the specific meaning of each command.

```

while(1)
{
    if(times>=25)//The timer accumulates every 100ms, and when it reaches 25 times, that
is, every 2.5 seconds, the state of the car is changed.
    {
        #if MOTOR_TYPE == 4
        Car_Move_PWM();
        #else
        Car_Move();
        #endif
        times = 0;
    }
    if(g_rcv_flag == 1)
    {
        g_rcv_flag = 0;

        #if UPLOAD_DATA == 1
        Deal_data_real();

        printf("M1:%d,M2:%d,M3:%d,M4:%d\r\n",Encoder_Now[0],Encoder_Now[1],Encoder_Now[2],Encoder_No
w[3]);

        #elif UPLOAD_DATA == 2
        Deal_data_real();

        printf("M1:%d,M2:%d,M3:%d,M4:%d\r\n",Encoder_Offset[0],Encoder_Offset[1],Encoder_Offset[2],E
ncoder_Offset[3]);

        #elif UPLOAD_DATA == 3
        Deal_data_real();

        printf("M1:%.2f,M2:%.2f,M3:%.2f,M4:%.2f\r\n",g_Speed[0],g_Speed[1],g_Speed[2],g_Speed[3]);
        #endif
    }
}

//The timer interrupt service function reads and prints data every 100ms
void TIMER_0_INST_IRQHandler(void)

```

```

{
    //If a timer interrupt occurs
    switch( DL_TimerG_getPendingInterrupt(TIMER_0_INST) )
    {
        case DL_TIMER_IIDX_ZERO://If it is 0 overflow interrupt
            times++;
            break;

        default:
            break;
    }
}

```

A 100ms timer is set in the program. Each time the timer interrupt is triggered, the times variable will be incremented by one. When it reaches 25 times, that is, 2.5 seconds, the motion state of the car will be changed. If the motor type is 4, that is, the motor without encoder, then the pwm version of the car state switching function is used. At the same time, read the data sent by the driver board and print the data out at the same time.

```

//Check the data sent from the driver board, and save the data that meets the communication
protocol
void Deal_Control_Rxtemp(uint8_t rxtemp)
{
    static u16 step = 0;
    static u8 start_flag = 0;

    if(rxtemp == '$' && start_flag == 0)
    {
        start_flag = 1;
        memset(g_recv_buff,0,RXBUFF_LEN);//Clear data
    }

    else if(start_flag == 1)
    {
        if(rxtemp == '#')
        {
            start_flag = 0;
            step = 0;
            g_recv_flag = 1;
            memcpy(g_recv_buff_deal,g_recv_buff,RXBUFF_LEN); //Only correct ones will be
assigned
        }
        else
        {
            if(step > RXBUFF_LEN)
            {
                start_flag = 0;
                step = 0;
                memset(g_recv_buff,0,RXBUFF_LEN);//Clear received data
            }
            else

```

```

        {
            g_rcv_buff[step] = rxtemp;
            step++;
        }
    }
}

//Format the data saved from the driver board and prepare it for printing
void Deal_data_real(void)
{
    static uint8_t data[RXBUFF_LEN];
    uint8_t length = 0;

    //Overall encoder
    if ((strcmp("MAll", (char*)g_rcv_buff_deal, 4) == 0))
    {
        length = strlen((char*)g_rcv_buff_deal) - 5;
        for (uint8_t i = 0; i < length; i++)
        {
            data[i] = g_rcv_buff_deal[i + 5]; //Remove the colon
        }

        data[length] = '\0';

        char* strArray[10]; //Pointer array The length is defined by the split number
        char 1 byte char* 4 bytes
        char mystr_temp[4][10] = {'\0'};
        splitString(strArray, (char*)data, ", "); //Split by comma
        for (int i = 0; i < 4; i++)
        {
            strcpy(mystr_temp[i], strArray[i]);
            Encoder_Now[i] = atoi(mystr_temp[i]);
        }
    }

    //10ms real-time encoder data
    else if ((strcmp("MTEP", (char*)g_rcv_buff_deal, 4) == 0))
    {
        length = strlen((char*)g_rcv_buff_deal) - 5;
        for (uint8_t i = 0; i < length; i++)
        {
            data[i] = g_rcv_buff_deal[i + 5]; //Remove the colon
        }

        data[length] = '\0';

        char* strArray[10]; //Pointer array The length is defined by the split number
        char 1 byte char* 4 bytes
        char mystr_temp[4][10] = {'\0'};
        splitString(strArray, (char*)data, ", "); //Split by comma
        for (int i = 0; i < 4; i++)
        {

```

```

        strcpy(mystr_temp[i],strArray[i]);
        Encoder_Offset[i] = atoi(mystr_temp[i]);
    }
}
//Speed
else if ((strncmp("MSPD",(char*)g_rcv_buff_deal,4)==0))
{
    length = strlen((char*)g_rcv_buff_deal)-5;
    for (uint8_t i = 0; i < length; i++)
    {
        data[i] = g_rcv_buff_deal[i+5]; //Remove the colon
    }

    data[length] = '\0';

    char* strArray[10];//Pointer array The length is defined by the split number
char 1 byte char* 4 bytes
    char mystr_temp[4][10] = {'\0'};
    splitString(strArray,(char*)data, ", ");//Split by comma
    for (int i = 0; i < 4; i++)
    {
        strcpy(mystr_temp[i],strArray[i]);
        g_Speed[i] = atof(mystr_temp[i]);
    }
}
}

```

- Deal_Control_Rxtemp: Filter the received data and save those that meet the communication protocol.
- Deal_data_real: Extract the saved original data and reconstruct a new print format.

1.3Experimental phenomenon

After the wiring is correct, write the program into the mainboard. After resetting, you can see that the car will move forward for 2.5S, move backward for 2.5S, rotate right for 2.5S, rotate left for 2.5S, then stop for 2.5S, and continue the above state switching actions. At the same time, you can see that the printed motor values are constantly changing in the serial port assistant.

M1:1118,M2:1127,M3:1136,M4:1119

M1:1118,M2:1127,M3:1136,M4:1119

M1:1118,M2:1127,M3:1136,M4:1119

M1:1118,M2:1127,M3:1136,M4:1119

M1:1118,M2:1127,M3:1136,M4:1119

M1:1118,M2:1127,M3:1136,M4:1119

M1:1118,M2:1127,M3:1136,M4:1119

M1:1118,M2:1127,M3:1136,M4:1119

M1:1118,M2:1127,M3:1136,M4:1119

M1:1118,M2:1127,M3:1136,M4:1119

M1:1118,M2:1127,M3:1136,M4:1119

M1:1118,M2:1127,M3:1136,M4:1119